

Lab 3: Component Modelling & Architectural Pattern Selection

Course: Software Engineering Lab

Project Name: Academic Elective Bidding & Allocation System (Key: AEB)

Problem Statement: PS #05 — Campus & Academic Operations

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Architecture Selection: We chose **Layered Architecture** for the Academic Elective Bidding & Allocation System.

1. Architectural Choice

The system is structured using a three-tier **Layered Architecture** comprising the Presentation Layer, Business Layer, and Data Layer.

2. Two Specific Scenario-Related Reasons

- **Decoupling Academic Bidding and Batch Allocation:** Course bidding involves real-time interactions with students to submit preferences and an offline batch solver by the Academic Registrar. In a layered model, the resource hungry constraint satisfaction engine is separated from user sessions, so the system won't lock up.
- **Centralized Business Rule Enforcement:** Hard-coded business rules like the maximum number of bidding credits (100) and checking for prerequisite courses are isolated into a separate Business Layer so that they cannot be bypassed through any of the client interfaces.

3. Security Advantage

- **Isolated Data Persistence & Role-Based Access:** The presentation components cannot query or update the database directly. All requests go through intermediate

business services to prevent data access by unauthorized parties, bid tampering and student access to competitor bid distributions.

4. Performance Benefit

- **Dedicated Compute for Constraint Solving:** By isolating the allocation solver in the business tier, the optimization algorithm can resolve timetable collisions and assign electives for 5,000+ students in under 30 seconds without degrading interactive portal response times.